

Pharmacology of Angina Pectoris

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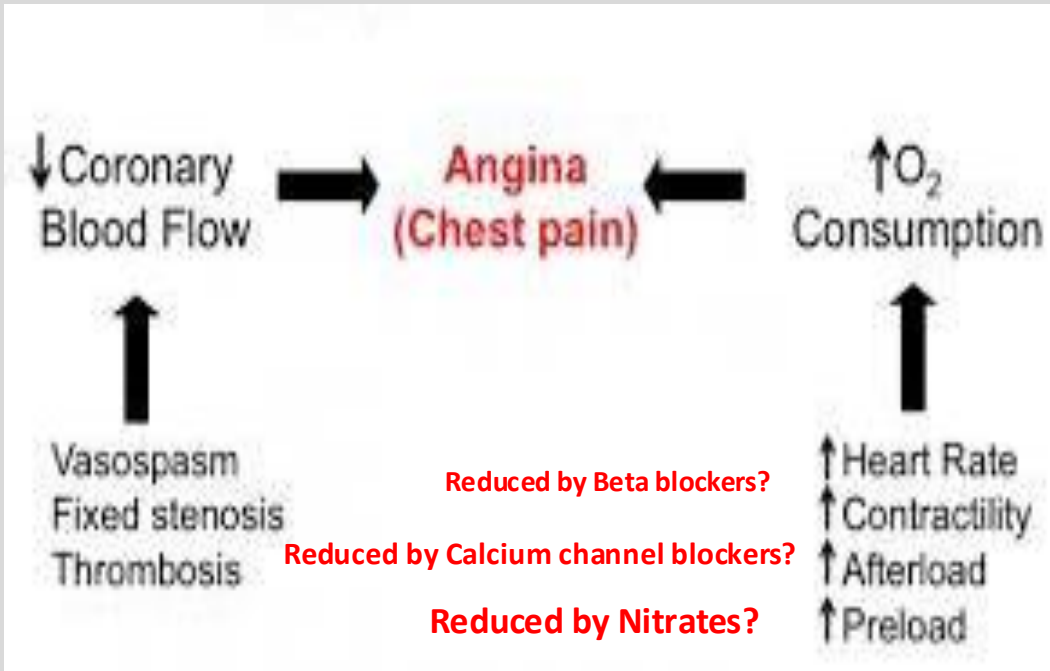
Session Objectives

- Review the pathophysiology of angina pectoris
- Distinguish between the types of anginas and the appropriate therapeutics.
- Describe the mechanism of nitrates in angina. Include adverse effects and specific drug interactions, mainly drugs for erectile dysfunction
- Review the mechanism of beta-adrenergic receptor antagonists. Identify patients which this class of drug is contraindicated.
- Describe the differences between the classes of calcium channel blockers.
- Identify other drugs used to manage angina, including ranolazine.
- Review the mechanisms of antiplatelet drugs.

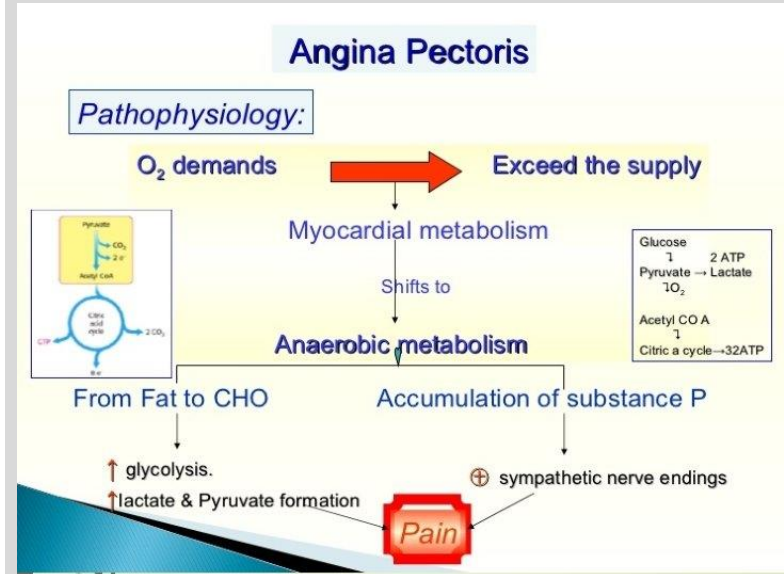
O₂ Supply and Demand Imbalance

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<https://www.cvphysiology.com/CAD/CAD007>



<https://slidetodoc.com/angina-angina-pectoris-learning-outcomes-on-completion-of/>

Angina and Drug Management

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+ Types of Angina

✓ Stable Angina

- "Effort" angina: Triggered by physical or mental exertion.
- Resolves with rest or nitrates.



✓ Unstable Angina

- New onset or worsening angina that is unpredictable.
- Rest/meds do not resolve.
- **Acute coronary syndrome:**
May lead to MI



✓ Vasospastic Angina

- Aka, Variant/Prinzmetal
- Occurs spontaneously, often at rest. Common at night & early morning (esp. w/ exercise).
- More common in women & smokers.
- Responds to nitrates; Calcium-channel blockers suppress (Beta-blockers do not)



✓ Microvascular Angina

- Angina due to coronary microvascular dysfunction or vasospasm.
- Accounts for chest pain in up to 50% of patients w/out obstructive epicardial CAD.
- Can occur with exertion or at rest.
- May respond less well to nitrates.
- Can be difficult to distinguish from epicardial angina. Diagnose with PET or CMR.



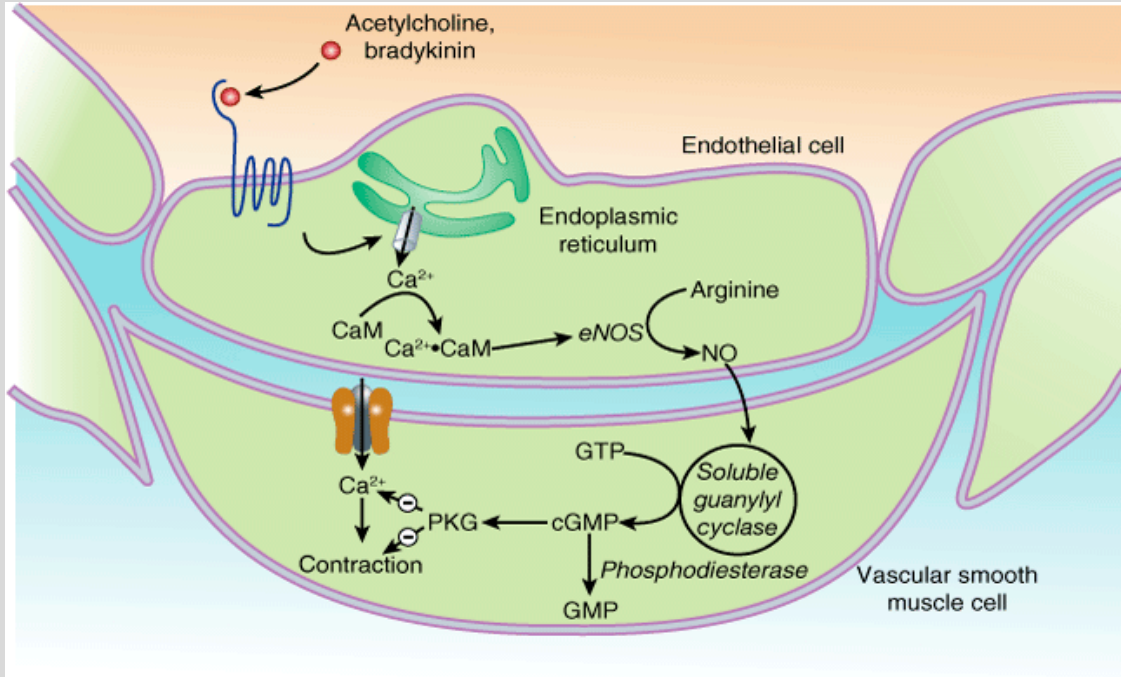
<https://www.drawittoknowit.com/course/pathology/glossary/pathophysiological-disorder/ischemic-heart-disease-symptoms-angina-others>

Stable: Nitrates, BB, CCB
Vasospastic: Nitrates, CCB
Unstable: Nitrates, CCB, BB **plus antiplatelets, ACE-, Statin. PCI.**

Nitrates: Increase cGMP

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Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology*, 11th Edition: <http://www.accessmedicine.com>

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Nitric oxide (NO) donors

- Nitroglycerin is denitrated by glutathione-S-transferase.
- Free nitrite is converted to NO.
- Another metabolic release of NO is by mitochondrial aldehyde reductase.
- Tolerance: occurs with continuous use. May be due to depletion of the sulfur group or inhibition of aldehyde reductase. An 8-hour drug holiday is required to restore action.
- NO donor results: Increased venous capacitance, decreased preload, reduced venous return = less cardiac output

Nitrates: Clinical Uses

- Immediate relief of angina: sublingual nitroglycerin
- Severe angina: IV nitroglycerin
- Recurrent typical and atypical angina: oral, buccal, transdermal
- Other uses: Heart Failure (decreases preload)
Unresponsive hypertension (sodium nitroprusside has added arteriolar vasodilator action).

Nitrates: Formulations and duration

- Nitroglycerin (sl): 10-30 min
- Isosorbide dinitrate (sl) 10-60 min
- Amyl nitrite inhaled 3-5 min
- Nitroglycerin (oral): 6-8 hr
- Nitroglycerin (transdermal): 8-10 hr
- Isosorbide dinitrate (oral, chewable): 2-6 hr
- Isosorbide mononitrate (oral; 100% bioavailable): 6-10 hr

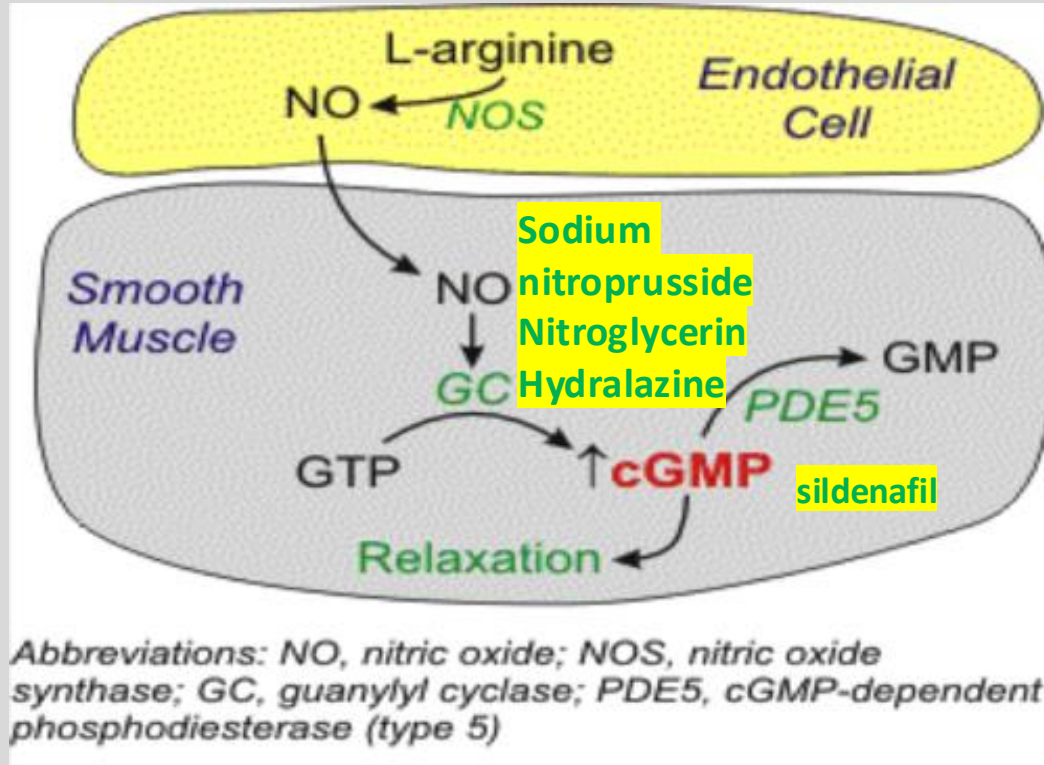
Nitrates: Adverse Effects

- Throbbing, vascular headache
- Vasodilation, hypotension (orthostatic) and compensatory tachycardia, with increased ventricular contractility
- Compensatory retention of salt and water due to enhanced renin-angiotensin II-aldosterone.
- NO activates guanylyl cyclase. Increased cGMP decreases platelet activation. Use as antiplatelet not established.
- Drug interactions: PDE5 inhibitors: sildenafil, vardenafil, tadalafil.

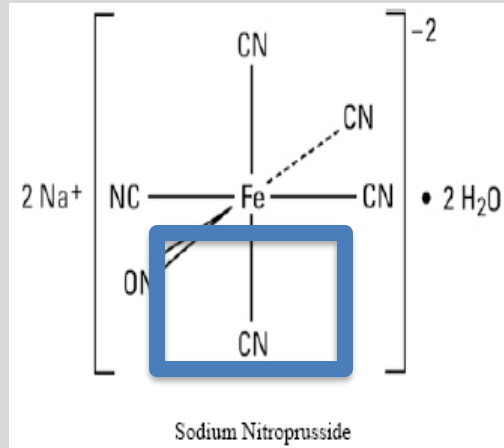
Nitrates and Erectile dysfunction drugs

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Sodium Nitroprusside



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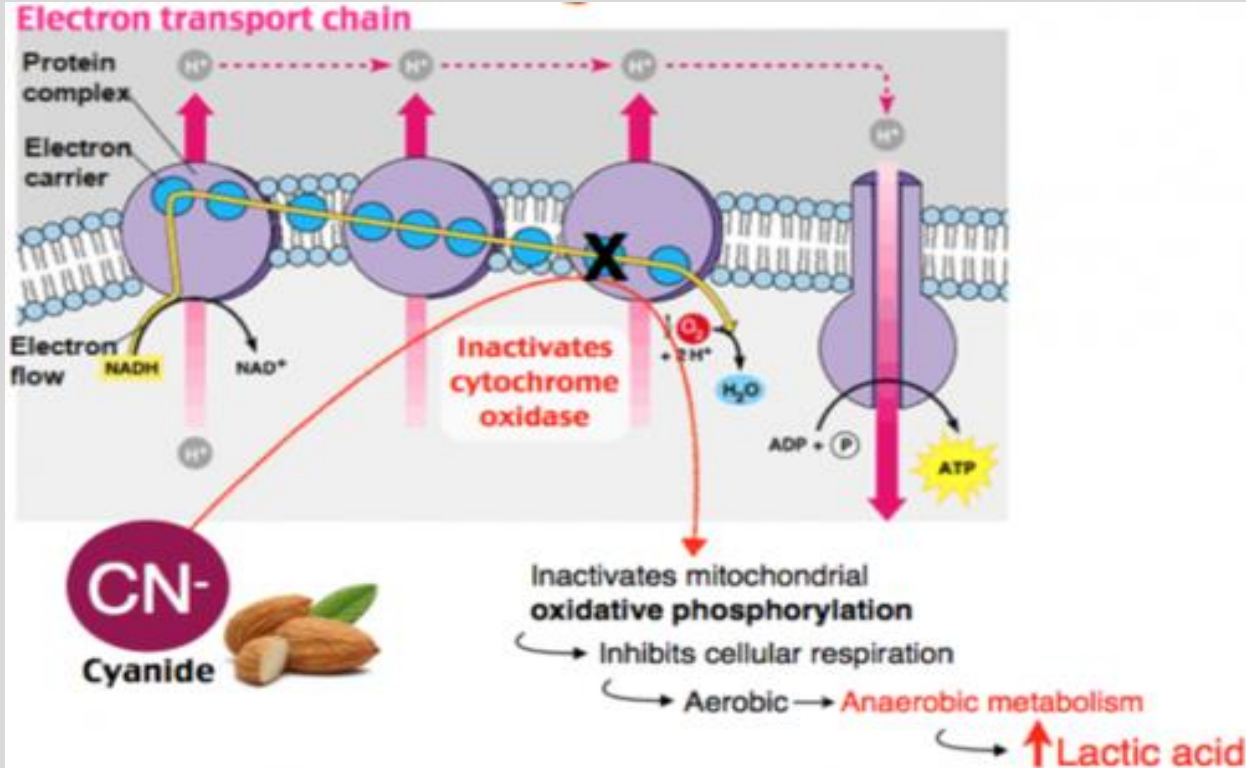
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- Short-acting NO donor, IV use for **hypertensive emergencies**. Dilates arterioles and venules. Decreases afterload and preload.
- Adverse effect: Rare accumulation of cyanide. Drug is a complex of iron, cyanide, and nitroso moiety.
- Antidote: sodium thiosulfate or hydroxycobalamin (vitamin B₁₂).

Cyanide Toxicity

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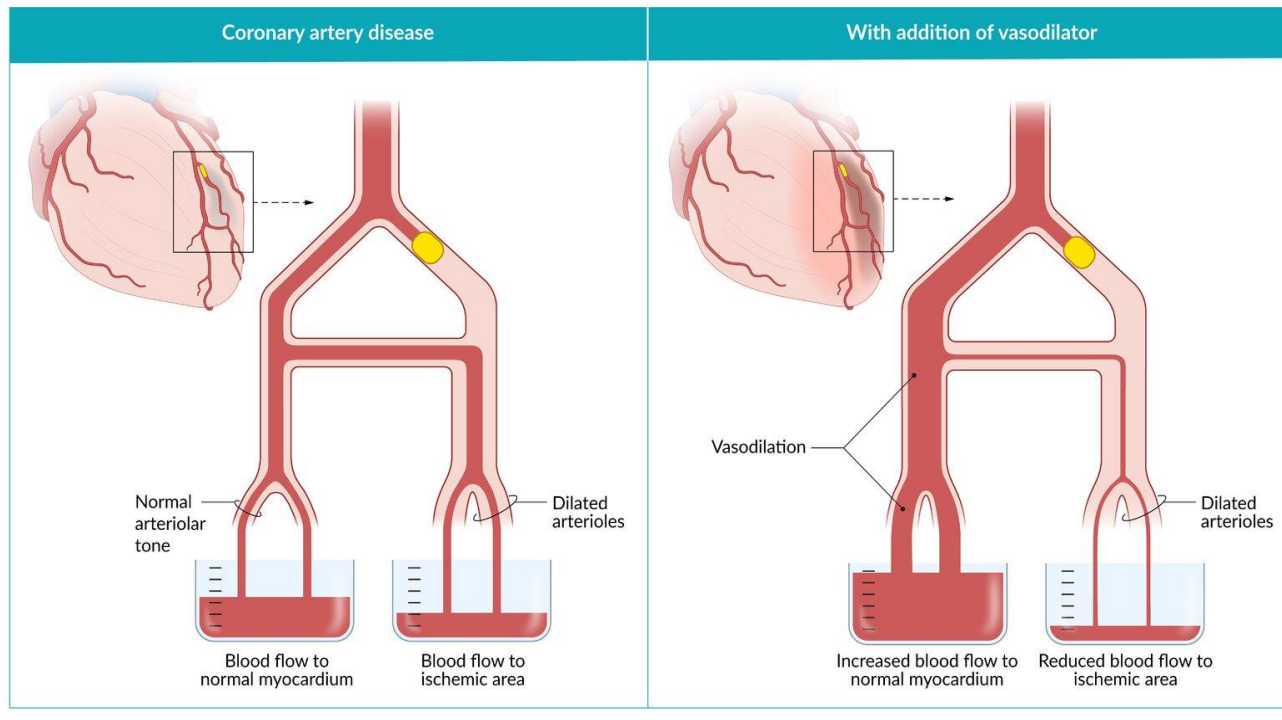
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Coronary Steal and Vasodilators

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<https://www.amboss.com/us/knowledge/coronary-artery-disease>

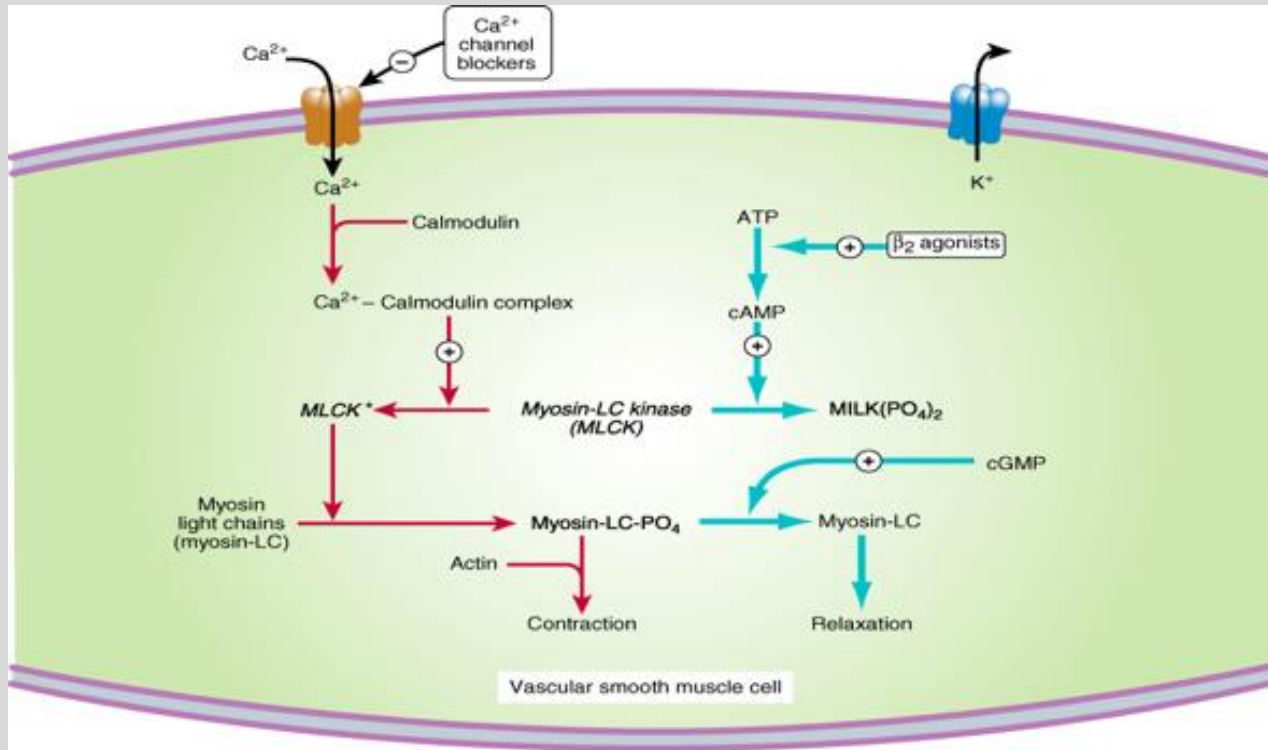
Calcium Channel Blockers (CCB)

- **Calcium is essential for the function of all nerves, smooth, cardiac and skeletal muscle, and exocrine glands.**
- **For CV, primary targets are arterioles and heart.**
- **Blockade of L-type channels**

Calcium Channel Blockers (CCB)

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Calcium Channel Blockers (CCB)

- **Arterioles relax, TPR and afterload decrease. Blood pressure falls, heart works less for systole. Common to all CCBs for antianginal and antihypertensive use.**
- **Verapamil and diltiazem: Heart automaticity and ventricular contractility are decreased. Class IV antiarrhythmic, antihypertensive, and antianginal use.**

Dihydropyridines (DHPs)

- Nifedipine types “dipines”
- Arteriole action only
- Decreases TPR, BP.
- May elicit the SANS reflex and tachycardia.
- **Amlodipine** and felodipine (once-a-day dosing)
- Nifedipine can be used in pregnant hypertension (first-line is methyldopa)

Verapamil and Diltiazem (non-DHP)

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- Arteriole and Heart action; Verapamil > Diltiazem
- Resist the SANS reflex, replaces beta blockers in COPD, asthma, diabetes patients
- Uses: hypertension, angina, Class IV antiarrhythmic (manages SVT, atrial fibrillation)
- Adverse Effects: Hypotension without high HR. SA and AV node depression. Negative inotropic effect = contraindicated in CHF.

CCB Adverse Effects

- Dizziness, headache, edema
- Tachycardia (DHP)
- Bradycardia (non-DHP)
- Constipation (most with verapamil)
- Gingival overgrowth (review past drugs??)
- Grapefruit drug interactions (CYP 3A4)
- Verapamil increases digoxin plasma levels.
- Antidote: Supportive care; IV calcium, insulin

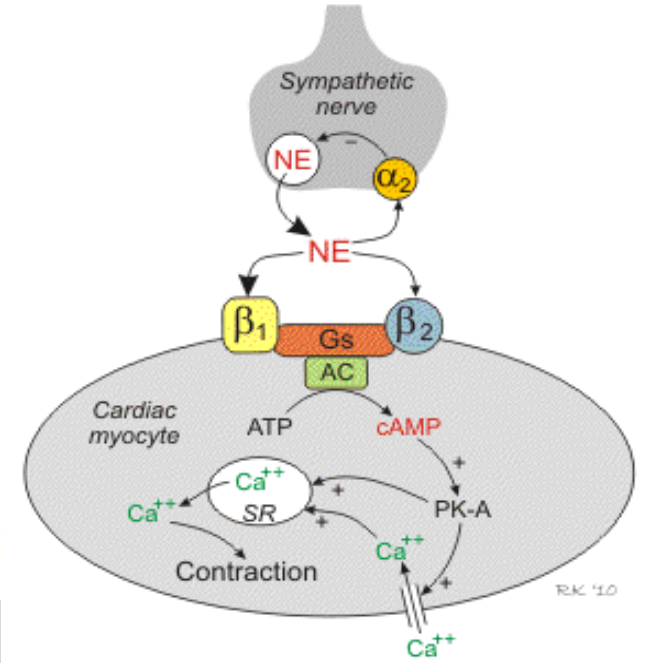
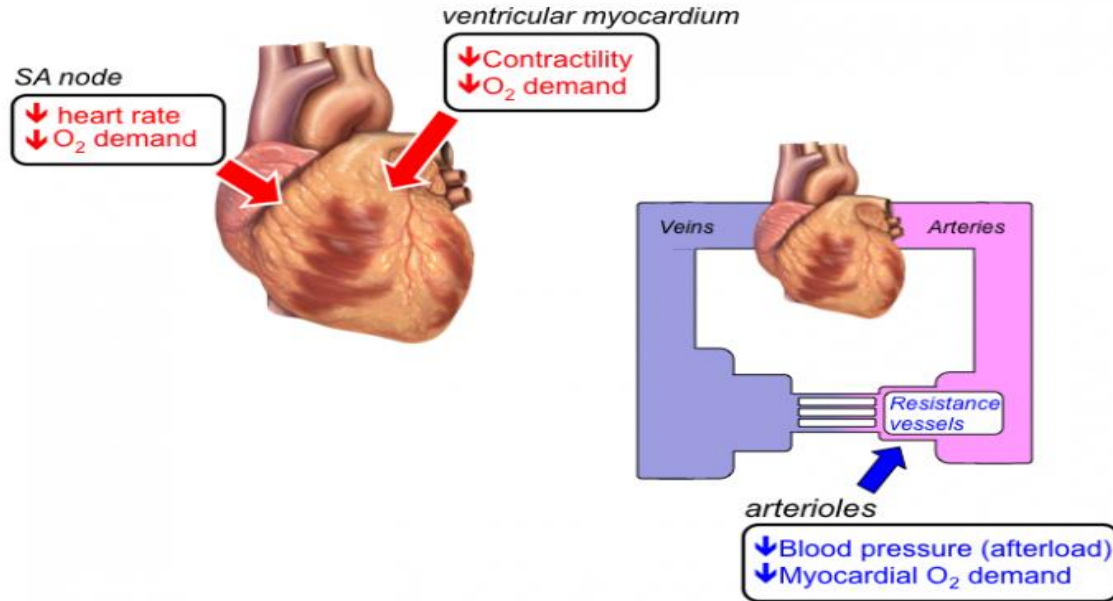


Beta-adrenergic receptor antagonists

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Beta Blockers



<https://www.cvpharmacology.com/cardioinhibitory/beta-blockers>

Abbreviations: NE, norepinephrine; Gs, G-stimulatory protein; AC, adenylyl cyclase; PK-A, cAMP-dependent protein kinase; SR, sarcoplasmic reticulum

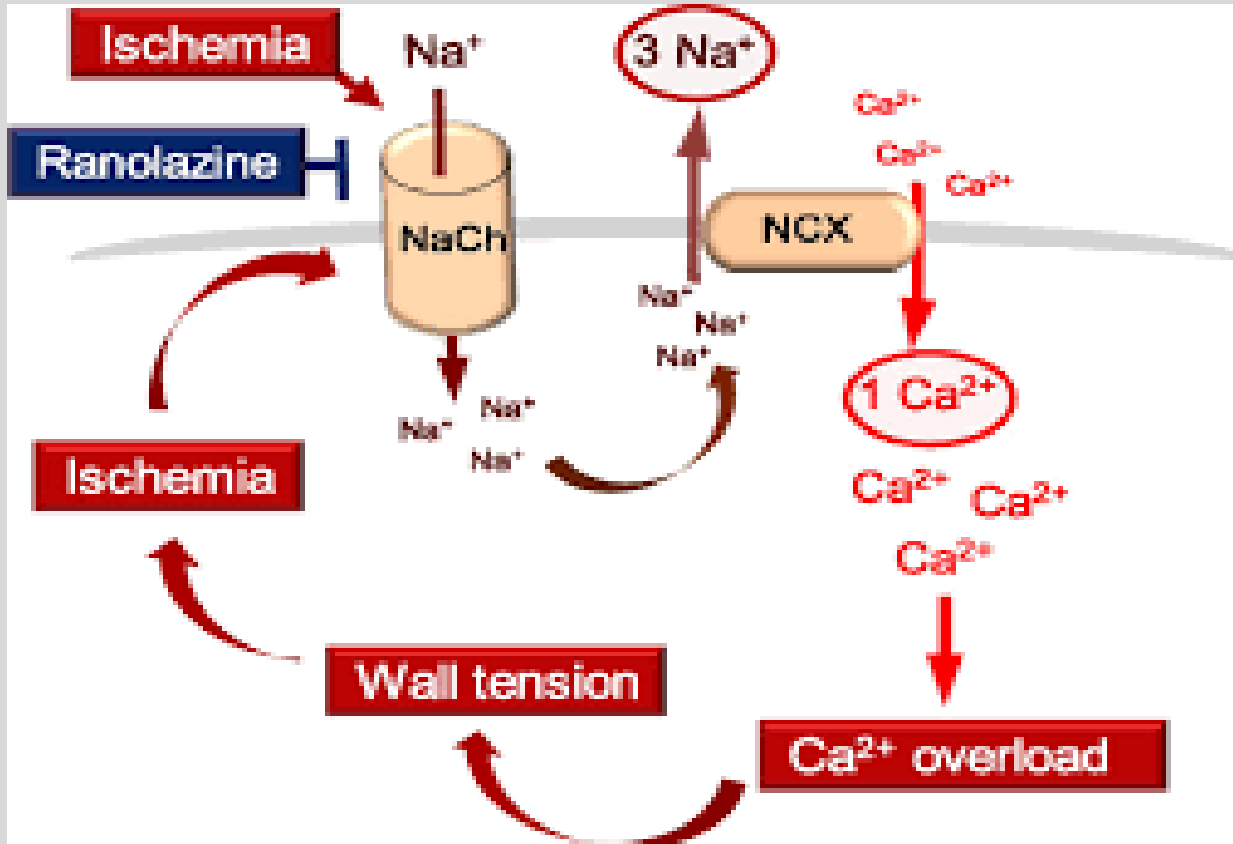
Beta-adrenergic receptor antagonists

- Beta 1: decrease HR, CO, and renin from JXA
- Use: hypertension, MI, CHF, antianginal (except vasospastic), Phase II antiarrhythmic
- Beta 1: (atenolol, metoprolol)—AV block most adverse effect
- Beta 1, 2: (nadolol, propranolol, timolol). Bronchoconstriction, hypoglycemia, complicates PVD (Beta 2)
- Labetalol and Carvedilol (alpha and beta blocker)
- Pindolol and Acebutolol (ISA)-can worsen angina
- Esmolol (ultrashort acting)
- Nebivolol (nitric oxide effect)
- Antidote: supportive therapy, glucagon

Ranolazine

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Ranolazine

- Reduces late sodium current. Prevents calcium entry via the $\text{Na}^+/\text{Ca}^{2+}$ exchanger.
- Angina prophylaxis (chronic)
- Adverse effects: nausea, constipation, dizziness.
CYP inhibitors increase ranolazine action.
Prolongs QT interval.

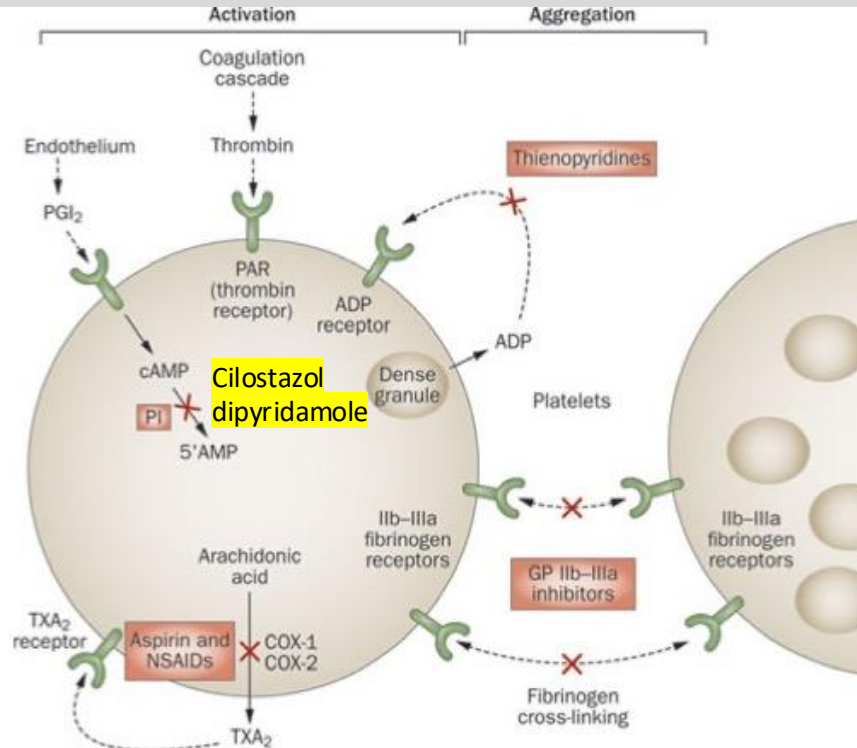
Other Therapies

- Percutaneous Coronary Intervention (PCI)
- Coronary artery bypass grafting (CABG)
- Antihypertensives (ACEIs)
- Smoking Cessation/Lifestyle modifications
- **Antiplatelet drugs (Review)**
- Trimetazidine (metabolic modifier. Not US approved)
- Lipid-lowering agents (future lecture)
- Ivabradine (future lecture)
- Antidiabetics (future lecture)

Antiplatelet Drugs

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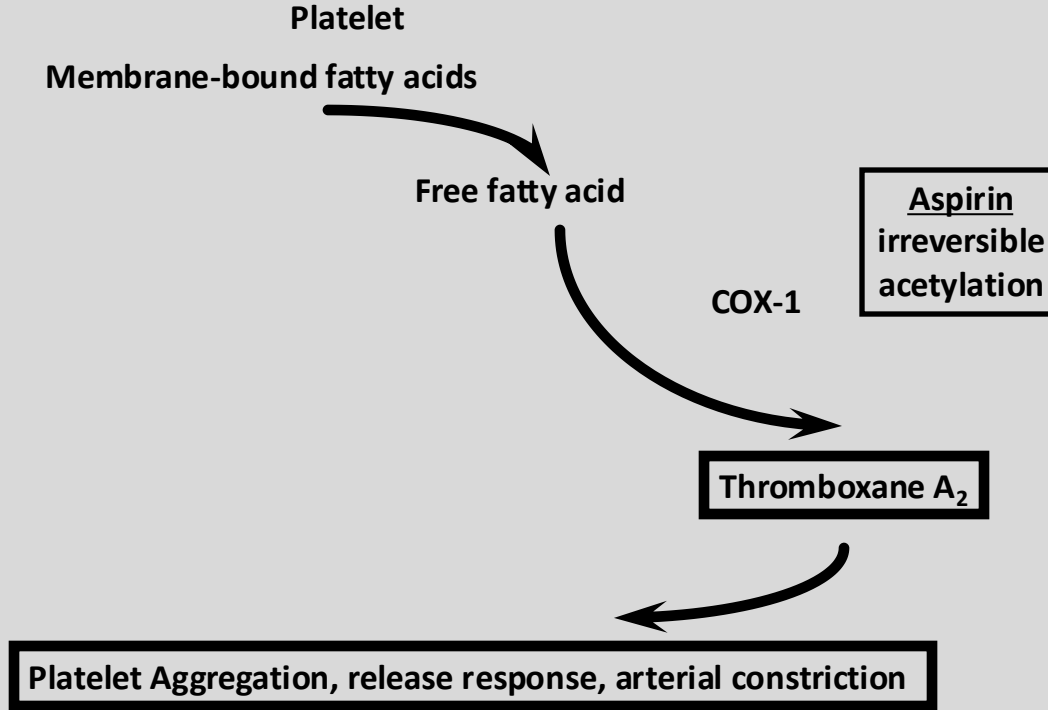
Antiplatelet drugs

- Aspirin **irreversibly** blocks COX 1,2 (NASIDs reversible). Analgesic, anti-inflammatory, antipyretic
- ADP receptor blockers (clopidogrel)
- Glycoprotein IIb/IIIa inhibitors (abciximab)
- Phosphodiesterase inhibitors (cilostazol, dipyridamole)—for peripheral artery disease

Thromboxane Synthesis

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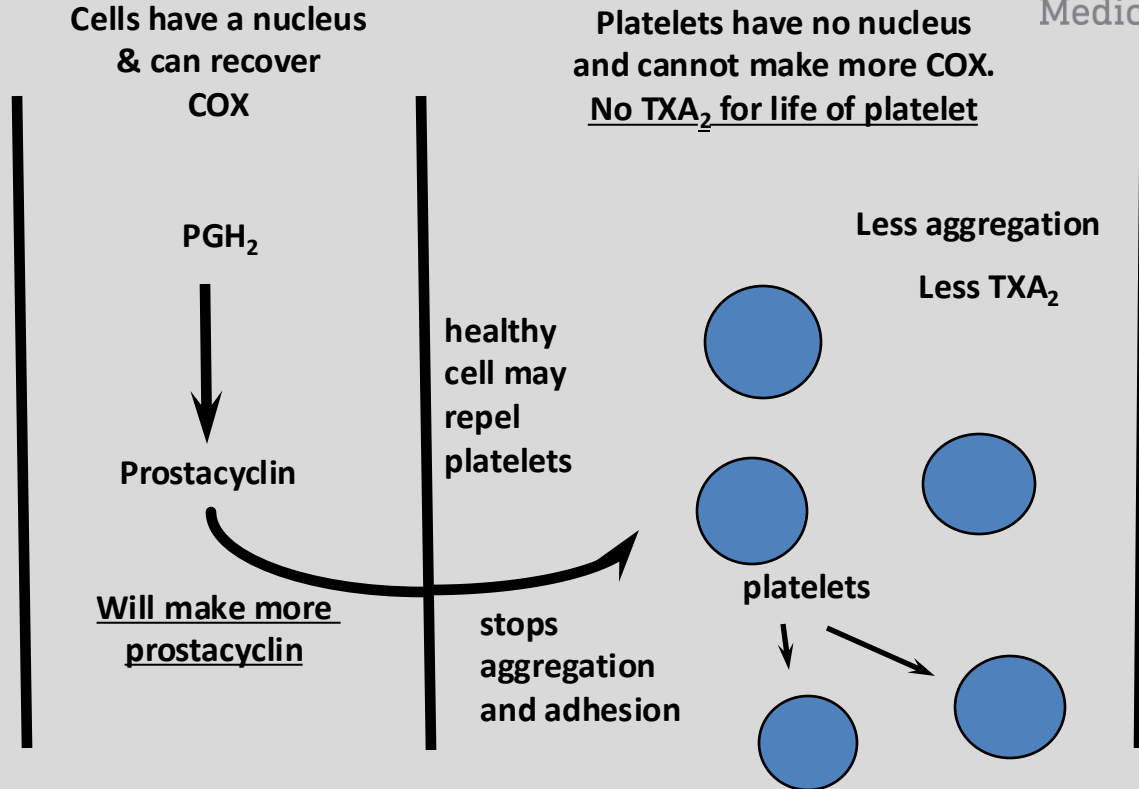
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Low Dose Aspirin Blocks TXA_2 not Prostacyclin

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Aspirin

- Prevents (primary) cardiovascular disease and recurrence (secondary)
- Consider individual patient (benefit vs risk)
- Adverse effects: GI bleed, hypersensitivity, metabolic acidosis (overdose). Uncoupling of oxidative phosphorylation.
- Salicylates not for pediatric patients.

ADP receptor blockers

- Ticlopidine, clopidogrel, prasugrel, ticagrelor, and cangrelor
- All are prodrugs, but ticagrelor and **cangrelor (IV)**.
- Clopidogrel is activated by 2C19 (mutations or drug inhibiting 2C19 will reduce antiplatelet effect).
- Clopidogrel: for unstable angina or non-ST-elevation acute myocardial infarction (NSTEMI) in combination with aspirin (also STEMI, recent MI, and PAD)
- All prevent thrombosis: Considered standard practice in patients undergoing stenting. And for acute coronary syndromes.
- Adverse effect: bleeding

Ticlopidine (Neutropenia)

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Ticlopidine Warning

Ticlopidine hydrochloride can cause Neutropenia/Agranulocytosis. Among 2048 patients in clinical trials, there were 50 cases (2.4%) of neutropenia (less than 1200 neutrophils/mm³), and the neutrophil count was below 450/mm³ in 17 of these patients (0.8% of the total population).

Life-threatening hematological adverse reactions, including neutropenia/agranulocytosis, thrombotic thrombocytopenic purpura (TTP) and aplastic anemia have occurred.

The incidence of TTP peaks after about 3 to 4 weeks of therapy and neutropenia peaks at approximately 4 to 6 weeks. The incidence of aplastic anemia peaks after about 4 to 8 weeks of therapy. The incidence of the hematologic adverse reactions declines thereafter. Only a few cases of neutropenia, TTP, or aplastic anemia have arisen after more than 3 months of therapy.

GP IIb/IIIa Inhibitors

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- **Abciximab (monoclonal antibody)-No longer available in U.S.**
- **Reversible inhibitors: Eptifibatide, tirofiban. Short half lives= administer by continuous infusion.**
- **Use: Acute coronary syndrome or PCI in those with higher thrombus risk, unable to tolerate ADP receptor blockers, or aspirin allergy.**
- **Adverse effect: bleeding**
- **Pathology Correlate: Glanzmann disease results from absent or defective GP IIb/IIIa.**

Summary Slide

- **Distinguish between the types of anginas and the appropriate therapeutics.**
- **Identify other drugs used to manage angina, including ranolazine. Review the mechanisms of antiplatelets.**

Core References:

1. **Katzung & Trevor Basic and Clinical Pharmacology, 16e, 2024; Chapter 12: Vasodilators and Treatment of Angina Pectoris; Chapter 11: Antihypertensive Agents; Chapter 34: Drugs Used in Disorders of Coagulation.**
2. **Pharmacology of Angina Pectoris Summary Guide (Maria A. Pino, Ph.D)**
3. **Scholar Bricks (CPG)**

Lecture Feedback Form:

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>